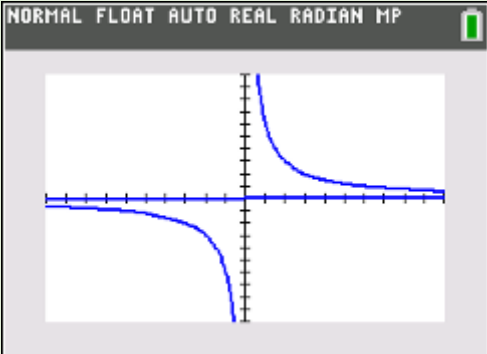
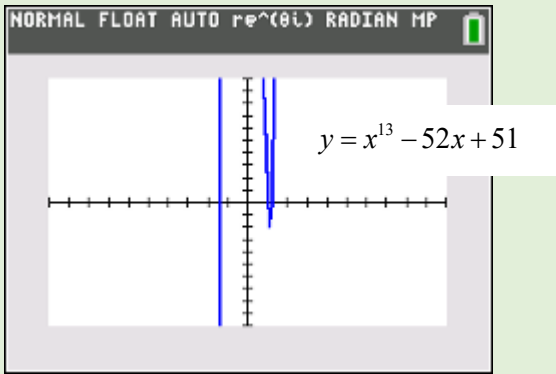
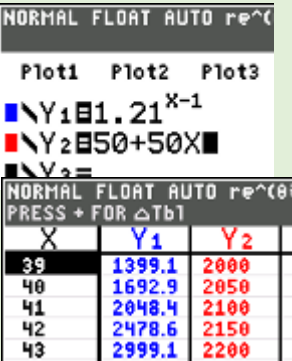
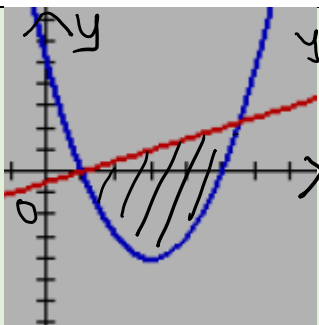


| Qn | Solutions                                                                                                                                                                                                                                                                                                                                                                                                                                | Comments                                                          |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| 1  | <p>A curve <math>C</math> has parametric equations</p> $x = \frac{3}{t}, \quad y = 2t.$ <p>(i) The line <math>y = 2x</math> cuts <math>C</math> at the points <math>A</math> and <math>B</math>. Find the exact length of <math>AB</math>. [3]</p>                                                                                                                                                                                       |                                                                   |
| 1i | <p>At intersection,</p> $2t = 2\left(\frac{3}{t}\right)$ $2t = \frac{6}{t}$ $t^2 = 3$ $t = \pm\sqrt{3}$ <p>The coordinates of <math>A</math> and <math>B</math> are <math>(\sqrt{3}, 2\sqrt{3})</math> and <math>(-\sqrt{3}, -2\sqrt{3})</math> respectively.</p> $AB = \sqrt{(\sqrt{3} + \sqrt{3})^2 + (2\sqrt{3} + 2\sqrt{3})^2}$ $= \sqrt{12 + 48}$ $= 2\sqrt{15}$ <p>Length of <math>AB</math> is <math>2\sqrt{15}</math> units.</p> | <p>This portion is from secondary school coordinate geometry.</p> |
|    | <p>(ii) The tangent at the point <math>P\left(\frac{3}{p}, 2p\right)</math> on <math>C</math> meets the <math>x</math>-axis at <math>D</math> and the <math>y</math>-axis at <math>E</math>. The point <math>F</math> is the midpoint of <math>DE</math>. Find a cartesian equation of the curve traced by <math>F</math> as <math>p</math> varies. [5]</p>                                                                              |                                                                   |
| ii | $x = \frac{3}{t}, \quad y = 2t$ $\frac{dx}{dt} = -\frac{3}{t^2}, \quad \frac{dy}{dt} = 2$ $\frac{dy}{dx} = -\frac{2t^2}{3}$ <p>At <math>P\left(\frac{3}{p}, 2p\right), \frac{dy}{dx} = -\frac{2p^2}{3}</math></p> $y - 2p = -\frac{2p^2}{3}\left(x - \frac{3}{p}\right)$ $= -\frac{2p^2}{3}x + 2p$ $y = -\frac{2p^2}{3}x + 4p$ <p>When <math>y = 0, x = \frac{6}{p}</math>.</p>                                                          |                                                                   |

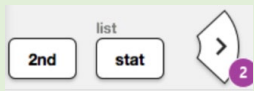
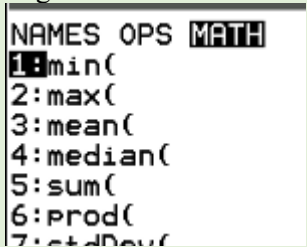
| Qn | Solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    | <p>When <math>x = 0, y = 4p</math>.</p> <p>Midpoint of <math>DE</math> is <math>\left(\frac{1}{2}\left(0 + \frac{6}{p}\right), \frac{1}{2}(4p + 0)\right) = \left(\frac{3}{p}, 2p\right)</math></p> <p>Coordinates of <math>F</math> are <math>\left(\frac{3}{p}, 2p\right)</math>.</p> <p><math>\therefore x = \frac{3}{p}, y = 2p</math></p> <p><math>p = \frac{y}{2}</math></p> <p><math>x = \frac{3}{\frac{y}{2}} = \frac{6}{y}</math></p> <p>Cartesian equation of curve traced out by <math>F</math> is <math>y = \frac{6}{x}</math>.</p> | <p>This portion is from secondary school coordinate geometry.</p> <p>We see <math>x = \frac{3}{p}, y = 2p</math> as parametric equation.</p> <p>For this portion, we are trying to obtain a cartesian equation from parametric equations. The strategy is trying to substitute <math>p</math>, so that we can express the equation in terms of <math>x</math> and <math>y</math>.</p> <p>Notice that the equation is the same as the equation of curve <math>C</math>. This is because the graph looks like this.</p>  |
| 2i | <p>An arithmetic progression has first term 3. The sum of the first 13 terms of the progression is 156.</p> <p>(i) Find the common difference. [2]</p>                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 2i | <p>AP: <math>a = 3, S_{13} = 156, d = ?</math></p> <p><math>156 = \frac{13}{2}(2(3) + 12d)</math></p> <p><math>d = \frac{3}{2}</math></p>                                                                                                                                                                                                                                                                                                                                                                                                       | <p>Use the formula for su of <math>n</math> terms of an AP.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

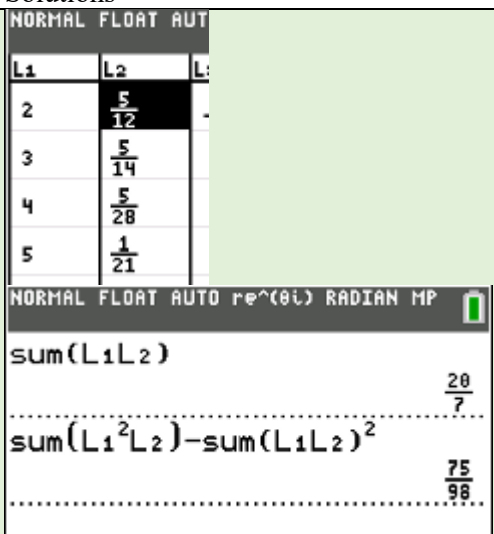
| Qn  | Solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                  |            |    |        |      |    |        |      |                                                                                                                                                                      |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|------------|----|--------|------|----|--------|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2ii | <p>A geometric progression has first term 3 and common ratio <math>r</math>. The sum of the first 13 terms of the progression is 156.</p> <p>(ii) Show that <math>r^{13} - 52r + 51 = 0</math>. Show that the common ratio cannot be 1 even though <math>r = 1</math> is a root of this equation. Find the possible values of the common ratio. [4]</p>                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                  |            |    |        |      |    |        |      |                                                                                                                                                                      |
| ii  | <p>GP: <math>a = 3, S_{13} = 156, r = ?</math></p> $\frac{3(1-r^{13})}{1-r} = 156$ $1-r^{13} = 52-52r$ $r^{13} - 52r + 51 = 0$ <p>If <math>r = 1, S_{13} = \underbrace{3+3+\dots+3}_{13 \text{ times}} = 39 \neq 156</math></p> <p>Using GC:</p> $r = -1.4511 \text{ or } r = 1 \text{ (NA) or } 1.2100$  <p>Possible values of <math>r</math> are -1.45 or 1.21</p>                                                                                                                                                          | <p>For solving this equation, we would be using a graphical approach. The polysm1t app can only solve up till a maximum degree of 10.</p> <p>Ask yourself: What's wrong if <math>r = 1</math>?</p> <p>For students who prefer more rigour, you might want to check that the graph <math>y = r^{13} - 52r + 51</math> has only 2 stationary points, so if we can find three roots on the graph (remember that <math>r = 1</math> is not an answer), we have found all the possible roots.</p> |                  |            |    |        |      |    |        |      |                                                                                                                                                                      |
| 2   | <p>(iii) It is given that the common ratio of the geometric progression is positive, and that the <math>n</math>th term of this geometric progression is more than 100 times the <math>n</math>th term of the arithmetic progression. Write down an inequality, and hence find the smallest possible value of <math>n</math>. [3]</p>                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                  |            |    |        |      |    |        |      |                                                                                                                                                                      |
| iii | <p>Since <math>r &gt; 0, r = 1.2100</math></p> $3(1.2100)^{n-1} > 100 \left( 3 + (n-1) \left( \frac{3}{2} \right) \right)$ $(1.2100)^{n-1} > 100 + 50n - 50$ $(1.2100)^{n-1} > 50 + 50n$ <p>By GC,</p> <table border="1"> <thead> <tr> <th><math>n</math></th><th><math>(1.2100)^{n-1}</math></th><th><math>50 + 50n</math></th></tr> </thead> <tbody> <tr> <td>41</td><td>2048.4</td><td>2100</td></tr> <tr> <td>42</td><td>2478.6</td><td>2150</td></tr> </tbody> </table> <p>smallest value of <math>n</math> is 42.</p>  | $n$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | $(1.2100)^{n-1}$ | $50 + 50n$ | 41 | 2048.4 | 2100 | 42 | 2478.6 | 2150 | <p>We can use the table method because the value of <math>n</math> that we are looking for is an integer. Alternatively, we could also use a graphical approach.</p> |
| $n$ | $(1.2100)^{n-1}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | $50 + 50n$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                  |            |    |        |      |    |        |      |                                                                                                                                                                      |
| 41  | 2048.4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2100                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                  |            |    |        |      |    |        |      |                                                                                                                                                                      |
| 42  | 2478.6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2150                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                  |            |    |        |      |    |        |      |                                                                                                                                                                      |

| Qn  | Solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Comments                                                                                                                 |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| 3   | <p>(a) The curve <math>y = f(x)</math> cuts the axes at <math>(a, 0)</math> and <math>(0, b)</math>. It is given that <math>f^{-1}(x)</math> exists. State, if it is possible to do so, the coordinates of the points where the following curves cut the axes.</p> <p>(i) <math>y = f(2x)</math></p> <p>(ii) <math>y = f(x - 1)</math></p> <p>(iii) <math>y = f(2x - 1)</math></p> <p>(iv) <math>y = f^{-1}(x)</math></p>                                                                                                                                                                                                                    | [4]                                                                                                                      |
| 3ai | <p>(Optional: <math>x</math> is replaced by <math>2x</math>, so the curve is stretched parallel to the <math>x</math>-axis, factor <math>\frac{1}{2}</math>.)</p> <p>The curve <math>y = f(2x)</math> cuts the axes at <math>\left(\frac{a}{2}, 0\right)</math> and <math>(0, b)</math>.</p>                                                                                                                                                                                                                                                                                                                                                 | This line is optional to write in your solution but it is useful working.                                                |
| ii  | <p>(Optional: <math>x</math> is replaced by <math>x - 1</math>, so the curve is translated 1 unit in the positive <math>x</math>-direction.)</p> <p>The curve <math>y = f(x - 1)</math> cuts the <math>x</math>-axis at <math>(a + 1, 0)</math>.</p>                                                                                                                                                                                                                                                                                                                                                                                         | We do not have a way to determine the $y$ -intercept, which is okay, because question says “if it is possible to do so”. |
| iii | <p>Method 1: from (i), replace <math>x</math> with <math>x - \frac{1}{2}</math>, so the curve in (i) is translated <math>\frac{1}{2}</math> units in the positive <math>x</math>-direction.</p> <p>The curve <math>y = f(2x - 1)</math> cuts the <math>x</math>-axis at <math>\left(\frac{a}{2} + \frac{1}{2}, 0\right)</math>.</p> <p>Method 2: From (ii), replace <math>x</math> with <math>2x</math>, so the curve in (ii) is stretched parallel to the <math>x</math>-axis, factor <math>\frac{1}{2}</math>.</p> <p>The curve <math>y = f(2x - 1)</math> cuts the <math>x</math>-axis at <math>\left(\frac{a+1}{2}, 0\right)</math>.</p> | $f(2x) \rightarrow f\left(2\left(x - \frac{1}{2}\right)\right)$ $= f(2x - 1)$                                            |
| iv  | The curve $y = f^{-1}(x)$ cuts the axes at $(0, a)$ and $(b, 0)$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Recall that for the graph of the inverse function, the $x$ - and $y$ - coordinates are reversed.                         |
| 3b  | <p>The function <math>g</math> is defined by</p> $g : x \mapsto 1 - \frac{1}{1 - x}, \text{ where } x \in \mathbb{R}, x \neq a.$ <p>(i) State the value of <math>a</math> and explain why this value has to be excluded from the domain of <math>g</math>.</p>                                                                                                                                                                                                                                                                                                                                                                               | [2]                                                                                                                      |
| bi  | <p><math>a = 1</math>.</p> <p>The function <math>g</math> is undefined for <math>x = 1</math>. Hence this value has to be excluded from the domain of <math>g</math>.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                          |
|     | (ii) Find $g^2(x)$ and $g^{-1}(x)$ , giving your answers in simplified form.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | [4]                                                                                                                      |

| Qn  | Solutions                                                                                                                                                                                                                                                                                                  | Comments                                                                                                                                                                                                                                                                                    |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ii  | $g^2(x) = g\left(1 - \frac{1}{1-x}\right)$ $= 1 - \frac{1}{1 - \left(1 - \frac{1}{1-x}\right)}$ $= 1 - \frac{1}{\frac{1}{1-x}}$ $= 1 - (1-x)$ $= x$ <p>Let <math>y = 1 - \frac{1}{1-x}</math>.</p> $y(1-x) = 1-x-1$ $y - xy = -x$ $x(y-1) = y$ $x = \frac{y}{y-1}$ $\therefore g^{-1}(x) = \frac{x}{x-1}.$ | <p>For this question, there is no need to find the domains of <math>g^2</math> and <math>g^{-1}</math>.</p> <p>For those who are looking for a shortcut, if <math>gg(x) = x</math>,</p> $g^{-1}gg(x) = g^{-1}(x)$ $g(x) = g^{-1}(x)$ <p>This shortcut is fully acceptable too.</p>          |
|     | <b>(iii)</b> Find the values of $b$ such that $g^2(b) = g^{-1}(b)$ . [2]                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                             |
| iii | $g^2(b) = g^{-1}(b)$ $b = \frac{b}{b-1}$ $b^2 - b = b$ $b(b-2) = 0$ $b = 0 \text{ or } 2$                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                             |
| 4a  | A flat novelty plate for serving food on is made in the shape of the region enclosed by the curve $y = x^2 - 6x + 5$ and the line $2y = x - 1$ . Find the area of the plate. [4]                                                                                                                           |                                                                                                                                                                                                                                                                                             |
| 4a  |  <p><math>y = x^2 - 6x + 5</math></p> <p><math>2y = x - 1</math></p> $x^2 - 6x + 5 = \frac{x-1}{2}$ $2x^2 - 13x + 11 = 0$ $x = 1 \text{ or } 5.5$                                                                       | <p>Good habit to sketch the graphs to identify the required region.</p> <p>Question did not ask for exact answer, so we can use GC for most of the working.</p> <p>Be careful to use <math>\frac{x-1}{2} - (x^2 - 6x + 5)</math> instead of <math>(x^2 - 6x + 5) - \frac{x-1}{2}</math></p> |

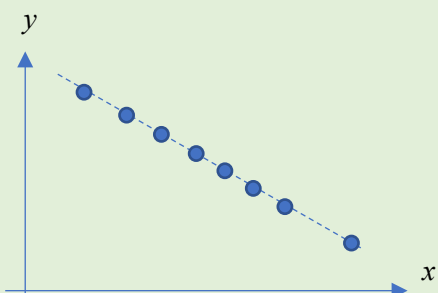
| Qn   | Solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Comments                                                                                                                                                                                                                                                       |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|      | $\text{Area of plate} = \int_1^{5.5} \frac{x-1}{2} - (x^2 - 6x + 5) dx$ $= 15.1875 \text{ or } \frac{243}{16} \text{ units}^2$                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                |
| 4b   | <p>A curved container has a flat circular top. The shape of the container is formed by rotating the part of the curve <math>x = \frac{\sqrt{y}}{a - y^2}</math>, where <math>a</math> is a constant greater than 1, between the points <math>(0, 0)</math> and <math>\left(\frac{1}{a-1}, 1\right)</math> through <math>2\pi</math> radians about the <math>y</math>-axis.</p> <p>(i) Find the volume of the container, giving your answer as a single fraction in terms of <math>a</math> and <math>\pi</math>. [4]</p> |                                                                                                                                                                                                                                                                |
| bi   | $\begin{aligned} \text{Volume of container} &= \pi \int_0^1 x^2 dy \\ &= \pi \int_0^1 y(a - y^2)^{-2} dy \\ &= -\frac{1}{2} \pi \int_0^1 -2y(a - y^2)^{-2} dy \\ &= -\frac{1}{2} \pi \left[ \frac{(a - y^2)^{-1}}{-1} \right]_0^1 \\ &= \frac{\pi}{2} \left( \frac{1}{a-1} - \frac{1}{a} \right) \\ &= \frac{\pi}{2} \left( \frac{1}{a(a-1)} \right) \\ &= \frac{\pi}{2a(a-1)} \text{ units}^3 \end{aligned}$                                                                                                            | <p>It is difficult to sketch this graph to visualize, so we have to rely on the description in the question.</p> <p>Read question, we need to simplify the answer into a single fraction.</p>                                                                  |
| 4bii | <p>(ii) Another curved container with a flat circular top is formed in the same way from the curve <math>x = \frac{\sqrt{y}}{b - y^2}</math> and the points <math>(0, 0)</math> and <math>\left(\frac{1}{b-1}, 1\right)</math>. It has a volume that is four times as great as the container in part (i). Find an expression for <math>b</math> in terms of <math>a</math>. [3]</p>                                                                                                                                      |                                                                                                                                                                                                                                                                |
| ii   | <p>Similar to (i),</p> $\text{Volume of second container} = \frac{\pi}{2b(b-1)} \text{ units}^3$ $\frac{\pi}{2b(b-1)} = 4 \left( \frac{\pi}{2a(a-1)} \right)$ $\frac{1}{b(b-1)} = \frac{4}{a(a-1)}$ $a(a-1) = 4b(b-1)$ $4b^2 - 4b - a(a-1) = 0$                                                                                                                                                                                                                                                                          | <p>Please do not redo the working you have done in (i) (just replace <math>a</math> with <math>b</math> to get <math>\frac{\pi}{2b(b-1)}</math>).</p> <p>Alternatively, you could solve this by completing the square, to make <math>b</math> the subject.</p> |

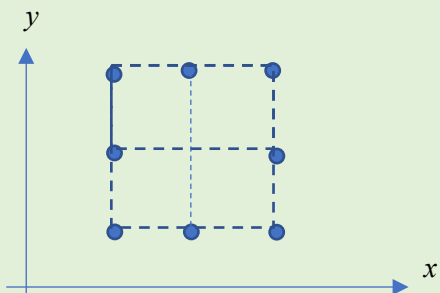
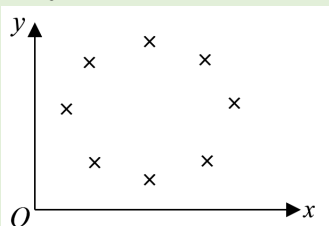
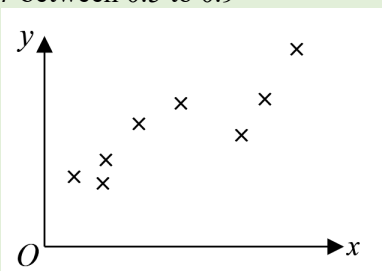
| Qn | Solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Comments                                                                                                                                                                                                                                                                                                                     |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    | $b = \frac{4 \pm \sqrt{16 + 16a(a-1)}}{8}$ $= \frac{4 \pm 4\sqrt{a^2 - a + 1}}{8}$ $= \frac{1 \pm \sqrt{a^2 - a + 1}}{2}$ <p>Since volume of container is formed in the same way, <math>b</math> is also greater than 1.</p> $\therefore b = \frac{1 + \sqrt{a^2 - a + 1}}{2}$                                                                                                                                                                                                                                                                                                                                                                                                          | <p>Need to reject one answer and we recall that <math>a</math>, <math>a</math> was greater than 1 in part ii.</p>                                                                                                                                                                                                            |
| 5i | <p>A bag contains 6 red counters and 3 yellow counters. In a game, Lee removes counters at random from the bag, one at a time, until he has taken out 2 red counters. The total number of counters Lee removes from the bag is denoted by <math>T</math>.</p> <p>(i) Find <math>P(T = t)</math> for all possible values of <math>t</math>. [3]</p>                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                              |
| 5i | $P(T = 2) = \frac{6}{9} \times \frac{5}{8} = \frac{5}{12}$ $P(T = 3) = 2! \left( \frac{6}{9} \times \frac{3}{8} \right) \times \frac{5}{7} = \frac{5}{14}$ $P(T = 4) = \frac{3!}{2!} \left( \frac{6}{9} \times \frac{3}{8} \times \frac{2}{7} \right) \times \frac{5}{6} = \frac{5}{28}$ $P(T = 5) = \frac{4!}{3!} \left( \frac{6}{9} \times \frac{3}{8} \times \frac{2}{7} \times \frac{1}{6} \right) \times \frac{5}{5} = \frac{1}{21}$ <div style="border: 1px solid black; padding: 5px; margin: 10px 0;"> <math display="block">\frac{3!}{2!} \left( \underbrace{\frac{6}{9} \times \frac{3}{8} \times \frac{2}{7}}_{2Y1R} \right) \times \underbrace{\frac{5}{6}}_R</math> </div> | <p>Recall that we need the last counter drawn to be red. We can vary the order of removal for the first <math>t - 1</math> counters. For e.g. for <math>P(T = 4)</math>, we need exactly 1 red and 2 yellow counter for his first 3 draws, then 1 final red count on his 4<sup>th</sup> draw.</p>                            |
| 5  | (ii) Find $E(T)$ and $\text{Var}(T)$ . [2]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                              |
| ii | <p>From GC,</p> $E(T) = \sum t P(T = t) = \frac{20}{7},$ $\text{Var}(T) = E(T^2) - [E(T)]^2$ $= \sum t^2 P(T = t) - [E(T)]^2$ $= \frac{75}{98}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <p>There is only 2 marks awarded for this question, so likely they are looking for GC methods. Press</p>  <p>to get to</p>  <p>To choose 5: sum(</p> |

| Qn  | Solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Comments                                                                                                                                      |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
|     |                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                               |
| 5   | <p>Lee plays this game 15 times.</p> <p>(iii) Find the probability that Lee has to take at least 4 counters out of the bag in at least 5 of his 15 games. [2]</p>                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                               |
| iii | $P(T \geq 4) = \frac{5}{28} + \frac{1}{21} = \frac{19}{84}$ <p>Let <math>X</math> be the number of games where Lee removes at least 4 counters, out of 15.</p> $X \sim B\left(15, \frac{19}{84}\right)$ $P(X \geq 5) = 1 - P(X \leq 4)$ $= 0.238$                                                                                                                                                                                                                                                       | <p>We are interested in “ number of times he takes at least 5 counters out of the bag in 15 games” – which is a binomial random variable.</p> |
| 6   | <p>A children’s game is played with 20 cards, consisting of 5 sets of 4 cards. Each set consists of a father, mother, daughter and son from the same family. The family names are Red, Blue, Green, Yellow and Orange. So, for example, the Red family cards are father Red, mother Red, daughter Red and son Red.</p> <p>The 20 cards are arranged in a row.</p> <p>(i) In how many different ways can the 20 cards be arranged so that the 4 cards in each family set are next to each other? [2]</p> |                                                                                                                                               |
| 6i  | Number of ways = $5! \times (4!)^5 = 955514880$                                                                                                                                                                                                                                                                                                                                                                                                                                                         | We have to take power 5 for 4! because there is arrangement in all the 5 families.                                                            |
| 6   | <p>(ii) In how many different ways can the cards be arranged so that all five father cards are next to each other, all four Red family cards are next to each other and all four Blue family cards are next to each other? [3]</p>                                                                                                                                                                                                                                                                      |                                                                                                                                               |

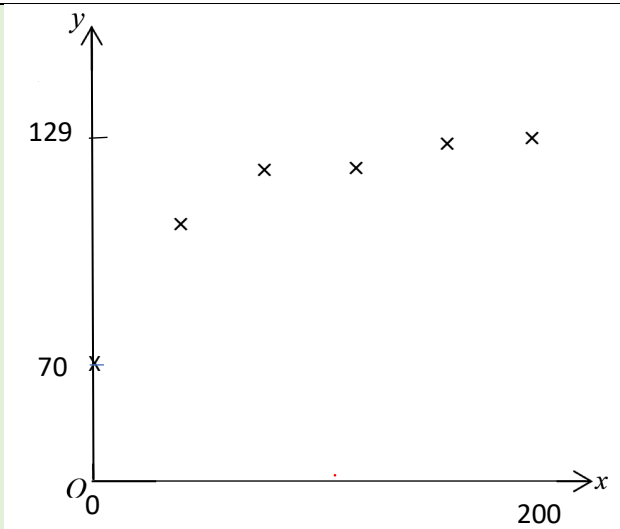


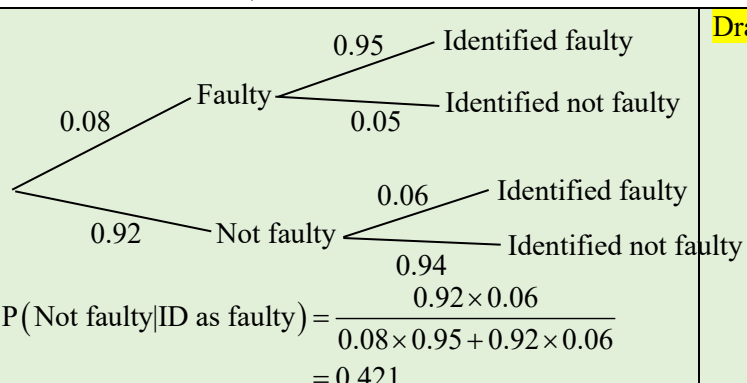
| Qn             | Solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Comments                                                                                                                                                                                                |             |                 |            |               |                                                                                                                                      |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-----------------|------------|---------------|--------------------------------------------------------------------------------------------------------------------------------------|
| ii             | <p>Hint: We need a unit that looks like this</p> <table><tr><td>Family of Blue</td><td>Father Blue</td><td>3 other fathers</td><td>Father Red</td><td>Family of Red</td></tr></table> <p>and 9 other family members.</p> <p>Number of ways = <math>10! \times 3! \times 2 \times (3!)^2 = 1567641600</math></p> <p>10!: arrange (9+1) units<br/>3!: arrange 3 other fathers<br/>2: group of blue and group of red can swap place<br/>3!^2: arrange 3 other “non-father” family of both blue and red</p> | Family of Blue                                                                                                                                                                                          | Father Blue | 3 other fathers | Father Red | Family of Red | <p>The families of Red and Blue can swap positions.</p> <p>The unit, together with the other 9 family members, will be 10 units.</p> |
| Family of Blue | Father Blue                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 3 other fathers                                                                                                                                                                                         | Father Red  | Family of Red   |            |               |                                                                                                                                      |
| 6              | <p>The cards are now arranged at random in a circle.</p> <p>(iii) Find the probability that no two father cards are next to each other. [4]</p>                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                         |             |                 |            |               |                                                                                                                                      |
| iii            | <p><math display="block">P(\text{no 2 fathers together}) = \frac{14! \times {}^{15}P_5}{19!}</math><math display="block">= \frac{1001}{3876} \text{ (or 0.258)}</math></p> <p>Probability question so you need to divide by the total – 19!</p>                                                                                                                                                                                                                                                         | <p>No 2 father cards together means that all the father cards are separated.</p> <p>- arrange the “non-father” cards<br/>- slot in the father cards into the gap to ensure that they are separated.</p> |             |                 |            |               |                                                                                                                                      |
| 7              | <p>The production manager of a food manufacturing company wishes to take a random sample of a certain type of biscuit bar from the thousands produced one day at his factory, for quality control purposes. He wishes to check that the mean mass of the bars is 32 grams, as stated on the packets.</p> <p>(i) State what it means for a sample to be random in this context. [1]</p>                                                                                                                  |                                                                                                                                                                                                         |             |                 |            |               |                                                                                                                                      |
| 7i             | <p>A sample is random if every biscuit bar from the thousands produced on a day has an equal chance of being selected and that the selections are independent of each other.</p>                                                                                                                                                                                                                                                                                                                        | <p>You can refer to your Sampling Distribution notes, just before Example 1.</p>                                                                                                                        |             |                 |            |               |                                                                                                                                      |
| 7ii            | <p>The masses, <math>x</math> grams, of a random sample of 40 biscuit bars are summarised as follows.</p> $n = 40 \qquad \Sigma(x - 32) = -7.7 \qquad \Sigma(x - 32)^2 = 11.05$ <p>(ii) Calculate unbiased estimates of the population mean and variance of the mass of biscuit bars. [2]</p>                                                                                                                                                                                                           |                                                                                                                                                                                                         |             |                 |            |               |                                                                                                                                      |
| ii             | <p>Let <math>w = x - 32</math>.</p> $\bar{x} = \bar{w} + 32 = 32 - \frac{7.7}{40} = 31.8075$ $s_x^2 = s_w^2 = \frac{1}{39} \left( 11.05 - \frac{(-7.7)^2}{40} \right) = 0.24533 \approx 0.245$                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                         |             |                 |            |               |                                                                                                                                      |
| 7              | <p>(iii) Test, at the 1% level of significance, the claim that the mean mass of biscuit bars is 32 grams. You should state your hypotheses and define any symbols you use. [5]</p>                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                         |             |                 |            |               |                                                                                                                                      |
| iii            | <p>Let <math>X</math> g be the mass of a biscuit bar and <math>\mu</math> g be the population mean.</p>                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                         |             |                 |            |               |                                                                                                                                      |

| Qn  | Solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Comments                                                                                                                                  |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
|     | $H_0 : \mu = 32$<br>$H_1 : \mu \neq 32$<br>Level of Significance: 1%<br>Since $n = 40$ is large, by Central Limit Theorem, $\bar{X}$ is approximately normal.<br>Test Statistic: When $H_0$ is true, $Z = \frac{\bar{X} - 32}{S/\sqrt{n}} \sim N(0,1)$<br>approximately.<br>Computation: $p\text{-value} = 0.0140$<br>Conclusion: Since $p\text{-value} = 0.0140 > 0.01$ , $H_0$ is not rejected at 1% level of significance. There is insufficient evidence to conclude that the population mean mass of the biscuit bars is not 32g. |                                                                                                                                           |
| 7   | (iv) Explain why there is no need for the production manager to know anything about the population distribution of the masses of the biscuit bars.                                                                                                                                                                                                                                                                                                                                                                                     | [2]                                                                                                                                       |
| iv  | Since the sample size 40 is sufficiently large, by Central Limit Theorem, the sample mean mass of the biscuit bars is approximately normally distributed. Hence, there is no need for the manager to know anything about the population distribution of the masses of the biscuit bars.                                                                                                                                                                                                                                                | These are <b>wrong</b> answers:<br>- $X$ or mass of biscuit is normally distributed by CLT.<br>- CLT can be applied                       |
| 8a  | Draw separate scatter diagrams, each with 8 points, all in the first quadrant, which represent the situation where the product moment correlation coefficient between variables $x$ and $y$ is<br>(i) $-1$ ,<br>(ii) $0$ ,<br>(iii) between $0.5$ and $0.9$ .                                                                                                                                                                                                                                                                          | [3]                                                                                                                                       |
| 8ai | The scatter plots given in 8a are not unique (there could be other correct answers – check with your tutors if unsure).                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                           |
|     | $r = -1$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Read the question clearly, the eight points should be in the 1st quadrant. All the points need to lie on a line with a negative gradient. |
|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                           |

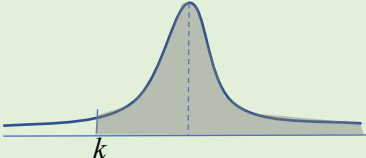
| Qn  | Solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Comments                                                                                                                                                                                                                                                                  |     |     |     |     |     |     |     |     |    |     |     |     |     |     |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|-----|-----|-----|-----|-----|-----|----|-----|-----|-----|-----|-----|
|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                           |     |     |     |     |     |     |     |     |    |     |     |     |     |     |
| ii  | $r = 0$<br>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | For $r = 0$ , we are aiming for <b>no LINEAR correlation</b> between the variables.                                                                                                                                                                                       |     |     |     |     |     |     |     |     |    |     |     |     |     |     |
| iii | $r$ between 0.5 to 0.9<br>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | For this scatter plot, the $y$ -coordinates of the points should not always be increasing as $x$ -coordinates increases, otherwise the $r$ -values might be too close to 1. However, most of the points need lie somewhat close to a line with <b>positive gradient</b> . |     |     |     |     |     |     |     |     |    |     |     |     |     |     |
| 8b  | <p>An investigation into the effect of a fertiliser on yields of corn found that the amount of fertiliser applied, <math>x</math>, resulted in the average yields of corn, <math>y</math>, given below, where <math>x</math> and <math>y</math> are measured in suitable units.</p> <table border="1" data-bbox="521 1355 1236 1456"><tr><td><math>x</math></td><td>0</td><td>40</td><td>80</td><td>120</td><td>160</td><td>200</td></tr><tr><td><math>y</math></td><td>70</td><td>104</td><td>118</td><td>119</td><td>126</td><td>129</td></tr></table> <p>(i) Draw a scatter diagram for these values. State which of the following equations, where <math>a</math> and <math>b</math> are positive constants, provides the most accurate model of the relationship between <math>x</math> and <math>y</math>.</p> <div style="display: flex; justify-content: space-around; align-items: flex-start;"><div style="text-align: center;"><p>(A) <math>y = ax^2 + b</math></p><p>(C) <math>y = a \ln 2x + b</math></p></div><div style="text-align: center;"><p>(B) <math>y = \frac{a}{x^2} + b</math></p><p>(D) <math>y = a\sqrt{x} + b</math></p></div></div> |                                                                                                                                                                                                                                                                           | $x$ | 0   | 40  | 80  | 120 | 160 | 200 | $y$ | 70 | 104 | 118 | 119 | 126 | 129 |
| $x$ | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 40                                                                                                                                                                                                                                                                        | 80  | 120 | 160 | 200 |     |     |     |     |    |     |     |     |     |     |
| $y$ | 70                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 104                                                                                                                                                                                                                                                                       | 118 | 119 | 126 | 129 |     |     |     |     |    |     |     |     |     |     |

[2]

| Qn  | Solutions                                                                                                                                                                                                                                                       | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| bi  |  <p>(D) <math>y = a\sqrt{x} + b</math> is the most accurate model.</p>                                                                                                         | <p>Take note that the point corresponding to (80,118) is only <u>slightly</u> lower than (120, 119).</p> <p>While there is no need to explain which model is the most accurate among the four models, it is good for us to know why. (B) and (C) are not accurate because <math>y</math> is not defined when <math>x = 0</math>. (A) is not accurate because as <math>x</math> increases, <math>y</math> increases at an increasing rate in the first quadrant (since <math>a</math> and <math>b</math> are both positive.)</p> |
| 8   | (ii) Using the model you chose in part (i), write down the equation for the relationship between $x$ and $y$ , giving the numerical values of the coefficients. State the product moment correlation coefficient for this model. [3]                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| ii  | <p>By GC, <math>y = 4.1821\sqrt{x} + 74.048</math></p> <p><math>y = 4.18\sqrt{x} + 74.0</math></p> <p><math>r = 0.981</math></p>                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 8   | (iii) Give two reasons why it would be reasonable to use your model to estimate the value of $y$ when $x = 189$ . [2]                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| iii | Since $x = 189$ is within the data range of $0 \leq x \leq 200$ and the product moment correlation coefficient $= 0.981$ is close to 1, it is reasonable to use the model to approximate the value of $y$ when $x = 189$ .                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 9i  | <p>On average 8% of a certain brand of kitchen lights are faulty. The lights are sold in boxes of 12.</p> <p>(i) State, in context, two assumptions needed for the number of faulty lights in a box to be well modelled by a binomial distribution. [2]</p>     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 9i  | <p>The probability that a light is faulty remains constant at 8%.</p> <p>Faulty lights occur independently of each other.</p> <div style="border: 1px solid black; padding: 5px; width: fit-content; margin: 10px auto;">Do you know why these are wrong?</div> | <p>Check your phrasing.</p> <p><b>Wrong</b> answers:</p> <ul style="list-style-type: none"> <li>- Probability of a light being faulty is independent ...</li> <li>- Prob. of a light being faulty in 12 lights is the same for each sample.</li> </ul>                                                                                                                                                                                                                                                                          |
| 9ii | <p>Assume now that the number of faulty lights in a box has a binomial distribution.</p> <p>(ii) Find the probability that a box of 12 of these kitchen lights contains at least 1 faulty light. [1]</p>                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| ii  | <p>Let <math>X</math> be the number of faulty lights, in a box of 12.</p> <p><math>X \sim B(12, 0.08)</math></p> <p><math>P(X \geq 1) = 1 - P(X = 0)</math></p> <p><math>= 0.63233</math></p> <p><math>\approx 0.632</math></p>                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

| Qn   | Solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Comments                                                                                                                                                                                    |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 9iii | The boxes are packed into cartons. Each carton contains 20 boxes.<br><br>(iii) Find the probability that each box in one randomly selected carton contains at least one faulty light. [1]                                                                                                                                                                                                                                                                  |                                                                                                                                                                                             |
| iii  | $P(\text{each box contains at least one faulty light})$<br>$= (0.63233)^{20}$<br>$= 0.000104$                                                                                                                                                                                                                                                                                                                                                              | Use at least 5sf in the answer from (ii). Alternatively, we can also solve this question using $B(20, 0.63233)$ .                                                                           |
| 9    | (iv) Find the probability that there are at least 20 faulty lights in a randomly selected carton. [2]                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                             |
| iv   | Let $Y$ be the number of faulty lights in a carton of 240 lights.<br>$Y \sim B(240, 0.08)$<br>$P(Y \geq 20) = 1 - P(Y \leq 19)$<br>$= 0.458$                                                                                                                                                                                                                                                                                                               | Each carton contains 20 boxes and each box contains 12 lights. $20 \times 12 = 240$ .                                                                                                       |
| 9    | (v) Explain why the answer to part (iv) is greater than the answer to part (iii). [1]                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                             |
| v    | The cases in (iii) is a <i>proper subset</i> of the cases in (iv).                                                                                                                                                                                                                                                                                                                                                                                         | For example, (iv) includes the case of 2 boxes contains 24 faulty lights, and there are 0 faulty lights in the other boxes. However, this is not included in (iii).                         |
| 9vi  | The manufacturer introduces a quick test to check if lights are faulty. Lights identified as faulty are discarded. If a light is faulty there is a 95% chance that the quick test will correctly identify the light as faulty. If the light is not faulty, there is a 6% chance that the quick test will incorrectly identify the light as faulty.<br><br>(vi) Find the probability that a light identified as faulty by the quick test is not faulty. [3] |                                                                                                                                                                                             |
| vi   |  $P(\text{Not faulty}   \text{ID as faulty}) = \frac{0.92 \times 0.06}{0.08 \times 0.95 + 0.92 \times 0.06} = 0.421$                                                                                                                                                                                                                                                   | Draw a tree diagram to help us.                                                                                                                                                             |
| 9    | (vii) Find the probability that the quick test correctly identifies lights as faulty or not faulty. [1]                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                             |
| vii  | $P(\text{correctly identifies the light})$<br>$= 0.08 \times 0.95 + 0.92 \times 0.94 = 0.9408$                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                             |
| 9    | (viii) Discuss briefly whether the quick test is worthwhile. [1]                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                             |
| viii | Answer 1<br>The quick test is not worthwhile, because from (vi), there is a 42% chance that a bulb identified as faulty by the test as faulty is actually not faulty.<br><br>Answer 2                                                                                                                                                                                                                                                                      | The purpose of this question is for us to use our answers in previous parts to support our decision. Depending on which part you decide to use, you will end up with different conclusions. |

| Qn   | Solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Comments                                                                                |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
|      | <p>The quick test is worthwhile, because from (vii), there is a 94% chance that the test correctly identifies a faulty bulb as faulty and a non-faulty bulb as non-faulty.</p> <p>(Not part of answer)<br/>           Note: In real life, to decide whether the test is worthwhile, we need to consider the cost (money, time, manpower, etc) incurred from running the quick test and the cost involved in wrongly discarding non-faulty lights.</p>                                                  |                                                                                         |
| 10i  | <p>A small component for a machine is made from two metal spheres joined by a short metal bar. The masses in grams of the spheres have the distribution <math>N(20, 0.5^2)</math>.</p> <p>(i) Find the probability that the mass of a randomly selected sphere is more than 20.2 grams. [1]</p>                                                                                                                                                                                                        |                                                                                         |
| 10i  | <p>Let <math>X</math> g be the mass of a sphere.<br/> <math>X \sim N(20, 0.5^2)</math><br/> <math>P(X &gt; 20.2) = 0.345</math></p>                                                                                                                                                                                                                                                                                                                                                                    |                                                                                         |
| 10ii | <p>In order to protect them from rusting, the spheres are given a coating which increases the mass of each sphere by 10%.</p> <p>(ii) Find the probability that the mass of a coated sphere is between 21.5 and 22.45 grams. State the distribution you use and its parameters. [3]</p>                                                                                                                                                                                                                |                                                                                         |
| ii   | <p><math>1.1X \sim N(22, 0.55^2)</math><br/> <math>P(21.5 &lt; 1.1X &lt; 22.45) = 0.612</math></p>                                                                                                                                                                                                                                                                                                                                                                                                     | $\text{Var}(1.1X) = 1.1^2 \text{Var}(X)$                                                |
| 10   | <p>(iii) The masses of the metal bars are normally distributed such that 60% of them have a mass greater than 12.2 grams and 25% of them have a mass less than 12 grams. Find the mean and standard deviation of the masses of metal bars. [4]</p>                                                                                                                                                                                                                                                     |                                                                                         |
| iii  | <p>Let <math>Y</math> g be the mass of a bar.<br/> <math>Y \sim N(\mu, \sigma^2)</math></p> $P(Y > 12.2) = 0.6 \qquad P(Y < 12) = 0.25$ $P\left(Z > \frac{12.2 - \mu}{\sigma}\right) = 0.6 \qquad P\left(Z < \frac{12 - \mu}{\sigma}\right) = 0.25$ $\frac{12.2 - \mu}{\sigma} = -0.25335 \qquad \frac{12 - \mu}{\sigma} = -0.67449$ $\mu - 0.25335\sigma = 12.2 \qquad \mu - 0.67449\sigma = 12$ <p>By GC, <math>\mu = 12.320 \approx 12.3</math> and <math>\sigma = 0.47490 \approx 0.475</math></p> | Use: PlySmlt2 to solve the simultaneous equations.                                      |
| 10   | <p>(iv) The probability that the total mass of a component, consisting of two randomly chosen coated spheres and one randomly chosen bar, is more than <math>k</math> grams is 0.75. Find <math>k</math>, stating the parameters of any distribution you use. [4]</p>                                                                                                                                                                                                                                  |                                                                                         |
| iv   | <p>Let <math>T</math> g be the total mass of 2 coated spheres and a bar.<br/> <math>T = 1.1X_1 + 1.1X_2 + Y</math></p>                                                                                                                                                                                                                                                                                                                                                                                 | <p>Find and write down the distribution of <math>T</math> first.</p> <p>For TI-84CE</p> |

| Qn | Solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    | $E(T) = 2(22) + 12.320 = 56.320$ $\text{Var}(T) = 2(0.55)^2 + (0.47490)^2 \approx 0.83053$ $T \sim N(56.320, 0.83053)$ <div data-bbox="715 360 975 472">Remember that<br/> <math display="block">\sigma = \sqrt{\text{Var}(T)}</math> </div> <div data-bbox="563 499 930 656">  </div> $P(T > k) = 0.75$ $k = 55.705 \approx 55.7$ <div data-bbox="304 689 975 788">Change to “&lt;” inequality before using invnorm function in GC. (Look at the graph above to see how to change.)</div> | <div data-bbox="995 230 1273 349"> <code>invNorm</code><br/> area: 0.75<br/> μ: 56.320<br/> σ: √(0.83053)<br/> Tail: LEFT CENTER RIGHT<br/> Paste </div> <div data-bbox="995 353 1225 383">Other earlier GC:</div> <div data-bbox="995 387 1225 517"> <code>invNorm</code><br/> area: .25<br/> μ: 56.320<br/> σ: √(0.83053)<br/> Paste </div> <div data-bbox="995 521 1225 689"> <p>*Must change to</p> <math display="block">P(T &gt; k) = 0.75</math> <math display="block">P(T &lt; k) = 0.25</math> <math display="block">k = 55.705 \approx 55.7</math> </div> |